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ASSIGNMENT

Critical Enquiry and Research Methodology

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1. Explain the Process of Identifying and Formulating a Research Problem

A research problem is a specific issue, difficulty, contradiction, or unanswered question that a researcher wants to investigate systematically. Identifying a suitable research problem is one of the most important stages of research because it determines the objectives, methodology, data collection, analysis, and expected outcomes of the study.

Process of Identifying and Formulating a Research Problem

1. Identify a Broad Area of Interest

The researcher first selects a general area based on academic interests, practical problems, professional experience, or current developments. For example, a computer science student may select cybersecurity as a broad area of interest.

2. Conduct a Preliminary Study

The researcher collects basic information about the selected area using textbooks, research papers, journals, conference papers, reports, and reliable online sources. This provides an initial understanding of what is already known.

3. Review Existing Literature

A detailed literature review is conducted to understand previous research, commonly used methods, findings, limitations, and unanswered questions. The researcher asks what has already been studied, which approaches have been used, and what limitations remain.

4. Identify the Research Gap

The researcher identifies an area that has not been sufficiently studied, has conflicting findings, or has an unresolved practical problem. For example, existing phishing detection systems may perform well on desktop websites but have limitations in detecting mobile-based phishing attacks.

5. Narrow Down the Problem

A broad topic should be converted into a specific and manageable problem. For example: Broad topic – Cybersecurity; Specific area – Phishing detection; Research problem – Improving phishing detection for mobile users using machine-learning techniques.

6. Define the Research Objectives

The objectives explain what the study intends to achieve. They should be clear, specific, measurable, and related to the research problem. For example, an objective may be to develop a machine-learning model for detecting suspicious mobile URLs.

7. Formulate Research Questions

Research questions convert the problem into questions that can be investigated. For example: How accurately can machine-learning techniques identify phishing URLs? Which features contribute most to detection performance?

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8. Assess Feasibility

The researcher checks whether the problem can realistically be studied considering available time, data, technology, skills, participants, and financial resources.

9. Formulate the Final Research Problem

Finally, the researcher writes a clear and concise statement describing the problem, its context, the gap in existing knowledge, and the purpose of the proposed study.

Characteristics of a Good Research Problem

- Clearly defined and specific.
- Relevant to the chosen field of study.
- Based on an identifiable research gap or practical issue.
- Researchable using appropriate data and methods.
- Feasible within the available time and resources.
- Ethically acceptable.
- Capable of producing meaningful results.

2. Differentiate Between Research Method and Research Methodology by Using Examples

Research method and research methodology are related but different concepts. A research method is a specific technique used to collect, process, or analyze data, whereas research methodology is the overall logical framework that explains why particular methods are selected and how the complete research will be conducted.

Basis	Research Method	Research Methodology
Meaning	A specific technique or procedure used to conduct a research activity.	The systematic framework and reasoning used to plan and conduct the research.
Focus	Focuses on how data is collected or analyzed.	Focuses on why particular methods are appropriate and how they fit into the complete study.
Purpose	Provides practical techniques for conducting research.	Provides the overall research strategy and justification.
Examples	Survey, interview, experiment, observation, statistical analysis, machine-learning algorithm.	Quantitative, qualitative, or mixed-method research design and the justification for selecting particular methods.
Scope	Usually a particular activity within research.	Covers the broader research design, philosophy, methods, analysis, and validation.
Question answered	How will the research activity be performed?	Why is this approach suitable, and how will the complete

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		research be conducted?

Example

Suppose a researcher wants to study student satisfaction with online learning. The research method may be an online questionnaire distributed to 200 students, followed by statistical analysis of the responses. The research methodology would explain why a quantitative approach was selected, why a survey is appropriate, how participants will be selected, how the questionnaire will be designed, how data will be analyzed, and how validity and reliability will be established.

3. What is a Research Gap? Explain Any Four Techniques Used to Identify a Research Gap from Existing Literature

A research gap is an aspect of a research topic that has not been adequately investigated, explained, or resolved by existing studies. A gap may occur because previous studies have limitations, contradictory findings, outdated data, insufficient populations or contexts, or because a new technology or situation has emerged.

Four Techniques for Identifying Research Gaps

1. Systematic Literature Review

The researcher systematically searches and examines a large number of relevant studies using predefined search terms, databases, and selection criteria. Comparing the findings helps reveal areas that have received insufficient attention. For example, a review of machine-learning-based phishing detection studies may reveal that most studies use desktop URL datasets while fewer examine mobile-specific datasets.

2. Analyze Limitations and Future Work

Research papers commonly contain sections describing limitations and recommendations for future research. Studying these sections can directly reveal problems that previous researchers were unable to address. For example, a paper may state that its model was tested on a small dataset and should be evaluated on a larger and more diverse dataset.

3. Compare Conflicting or Inconsistent Findings

Researchers can compare results from different studies to find contradictions or disagreements. If studies produce different conclusions under similar conditions, further investigation may be required to explain the difference. For example, two studies may report different levels of accuracy for the same machine-learning algorithm on different datasets.

4. Identify New Contexts, Populations, or Technologies

A research gap can be found by examining whether existing findings have been tested in new environments, populations, geographical regions, or technologies. A method developed for desktop users, for example, may not have been sufficiently evaluated for smartphones. The new context can provide an opportunity for further research.

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Other Indicators of a Research Gap

- Outdated datasets or technologies.
- Small or non-representative samples.
- Limited geographical or demographic coverage.
- Lack of comparative studies.
- Insufficient validation of existing models.
- Unresolved theoretical or practical problems.

4. What is a Hypothesis? Explain Null and Alternative Hypothesis

A hypothesis is a clear, testable, and tentative statement about the expected relationship between variables or about a possible explanation for a phenomenon. It is formulated from existing knowledge, theories, observations, or previous research and is tested using collected evidence.

Characteristics of a Hypothesis

- It should be clear and specific.
- It should be testable using empirical data.
- It should relate measurable variables.
- It should be logically derived from existing knowledge or literature.
- It should allow the researcher to accept, reject, or fail to reject a proposed relationship based on evidence.

Null Hypothesis (H_0)

The null hypothesis states that there is no significant relationship, difference, or effect between the variables being studied. It represents the default assumption that any observed difference is not statistically significant.

Example: A researcher wants to determine whether a new teaching method improves student examination scores.

H_0: The new teaching method does not produce a significant improvement in students' examination scores.

Alternative Hypothesis (H_1 or Ha)

The alternative hypothesis states that a significant relationship, difference, or effect exists between the variables. It represents the researcher's proposed explanation that is tested against the null hypothesis.

Example:

H_1: The new teaching method produces a significant improvement in students' examination scores.

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Difference Between Null and Alternative Hypotheses

Basis	Null Hypothesis (H_0)	Alternative Hypothesis (H_1/Ha)
Meaning	States that there is no significant effect, relationship, or difference.	States that a significant effect, relationship, or difference exists.
Role	Acts as the default assumption tested statistically.	Represents the proposed research effect or relationship.
Example	The treatment has no significant effect on performance.	The treatment has a significant effect on performance.

In hypothesis testing, statistical evidence is used to determine whether the null hypothesis can be rejected. Researchers normally avoid saying that the null hypothesis is 'proved true'; instead, they report whether there is sufficient evidence to reject it at a chosen significance level.

5. Design a Research Methodology Covering Data Collection, Preprocessing, Model Development, Evaluation, and Validation

A research methodology should provide a systematic plan for answering the research questions. The following methodology can be used for a machine-learning-based research project, such as developing a system to detect phishing websites.

Proposed Research Methodology

1. Research Design

A quantitative and experimental research design will be used. The study will collect structured data, preprocess it, develop one or more machine-learning models, and compare their performance using predefined evaluation metrics.

2. Data Collection

Relevant data will be collected from reliable and legally accessible sources. For a phishing detection study, the dataset may contain legitimate and phishing URLs along with features such as URL length, number of special characters, presence of an IP address, number of subdomains, HTTPS usage, domain information, and other relevant attributes. If primary data is required, it may be collected through surveys, experiments, interviews, or observations after obtaining appropriate consent.

3. Data Cleaning and Preprocessing

The collected data will be inspected for missing values, duplicate records, inconsistent formats, irrelevant attributes, and outliers. Duplicate records will be removed, missing values will be handled using appropriate techniques, categorical data will be encoded, numerical features may be normalized or standardized, and irrelevant features will be removed. Data preprocessing should be performed carefully to avoid data leakage.

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4. Exploratory Data Analysis

Statistical summaries and visualizations will be used to understand the distribution of variables, class balance, relationships between features, and potential anomalies. This stage helps determine whether additional preprocessing or feature engineering is necessary.

5. Feature Engineering and Selection

Useful features will be created or transformed from the raw data. Feature selection techniques may then be used to remove redundant or irrelevant variables. The selection process should be based only on information available in the training data to prevent leakage.

6. Model Development

The dataset will be divided into training and testing portions. Several suitable algorithms may be developed and compared, such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, or appropriate neural-network models. Hyperparameters can be tuned using the training data and cross-validation.

7. Model Evaluation

The developed model will be evaluated using appropriate metrics. For classification, common metrics include accuracy, precision, recall, F1-score, specificity, and ROC-AUC. A confusion matrix will also be used to examine true positives, true negatives, false positives, and false negatives.

8. Validation

Cross-validation will be used during model development to estimate how consistently the model performs on unseen data. After model selection is completed, a separate test set that was not used during training or tuning will be used for final validation. If possible, an external dataset can provide an additional check of generalization.

9. Comparison and Interpretation

The performance of different models will be compared using the selected evaluation metrics. The results will be interpreted in relation to the research questions and objectives rather than relying only on a single metric.

10. Reproducibility and Documentation

The dataset source, preprocessing steps, feature definitions, model parameters, evaluation metrics, software libraries, and experimental settings will be documented. Random seeds may be recorded where appropriate so that the experiment can be reproduced.

11. Ethical Considerations

The researcher will use data responsibly, respect privacy and consent requirements, protect sensitive information, and follow applicable institutional and legal requirements. Data sources and research limitations will be clearly reported.

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Research Methodology Flow

Research Problem → Literature Review → Research Gap → Research Questions/Hypothesis → Data Collection → Data Cleaning → Preprocessing → Exploratory Data Analysis → Feature Engineering → Model Development → Training and Tuning → Evaluation → Validation → Interpretation → Conclusions

Example of Evaluation Metrics

Metric	Purpose
Accuracy	Measures the proportion of all predictions that are correct.
Precision	Measures how many predicted positive cases are actually positive.
Recall	Measures how many actual positive cases are correctly identified.
F1-score	Provides a combined measure of precision and recall.
Specificity	Measures the ability to correctly identify negative cases.
ROC-AUC	Measures the model's ability to distinguish between classes across classification thresholds.

Conclusion

Research begins with identifying a clear and meaningful research problem and understanding the existing body of knowledge. A systematic literature review helps researchers identify research gaps and formulate research questions and hypotheses. Research methods are the specific techniques used to conduct research, whereas research methodology provides the overall reasoning and framework behind those methods. A well-designed methodology covering data collection, preprocessing, model development, evaluation, and validation helps ensure that research findings are systematic, reliable, and reproducible.